

Reinforcement Learning for Robust Text-to-Cypher Schema Selection

Hetionet Biomedical Graph Database

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Exam project report

Main Focus

Policy-gradient methods, reward design, hierarchical action spaces, schema-aware rewriting, and action masking for robust Text-to-Cypher schema selection.

Scientific Objective

The objective of this project is to study how Reinforcement Learning can control intermediate decisions in a Text-to-Cypher pipeline, instead of generating the final Cypher query directly.

The project investigates whether an RL agent can improve robustness to paraphrased biomedical questions by selecting rewriting strategies, estimating query complexity, and choosing the relevant Hetionet schema path.

Problem Description

The project focuses on Text-to-Cypher over Hetionet, a public biomedical graph database based on Neo4j.

Given a biomedical natural language question, the goal is to identify the sequence of Hetionet schema relations needed to build the corresponding Cypher query. The agent does not generate the full query token by token. Instead, it operates on intermediate structured decisions.

The problem is challenging because user questions may be written differently from the dataset style. For example: drugs -> compounds, genetic factors -> genes, medical condition -> disease, used for -> treats.

These paraphrases can cause the schema selector to choose wrong relations, especially in multi-hop questions.

The final pipeline uses Qwen 1.5B as a few-shot schema-aware rewriter, MiniLM embeddings, schema cue features, and a hierarchical Gymnasium environment. The agent first chooses a rewriting strategy, then chooses the hop count, and finally selects a schema-path candidate.

The reward is based on the agreement between the predicted schema path and the gold schema sequence extracted from the Cypher query. Exact match is used as a strict metric, while F1 gives partial credit for partially correct schema paths.

Tasks

1. Dataset and schema extraction. Load the Hetionet Text-to-Cypher dataset and extract the gold schema sequence from each Cypher query. Each schema sequence is represented as a sequence of Hetionet relations, for example Compound -TREATS_CtD-> Disease and Disease -PRESENTS_DpS-> Symptom.
2. Build a Gymnasium RL environment. Define a custom Gymnasium environment where the state includes the question embedding, rewriting information, hop-count information, and schema-aware features. The action space is hierarchical: rewriting strategy -> hop count -> schema path.
3. Implement query rewriting. Use Qwen 1.5B as an open-source few-shot rewriter. The agent can choose among different rewriting strategies: keep original, entity normalization, relation normalization, full normalization, and aggressive rewriting. A safety filter rejects rewrites that change the meaning of the question or modify important biomedical relations.
4. Design the reward function. Define a reward based on exact match and F1 between the predicted schema path and the gold schema path. The reward also considers hop-count correctness and penalizes invalid or incoherent decisions.
5. Train and compare policies. Train the policy using supervised warm-start and policy-gradient methods. REINFORCE with baseline and Actor-Critic variants are implemented and tested as additional RL approaches. The best final configuration is the warm-started hierarchical schema-path policy with schema cue features and lexical action masking.
6. Improve multi-hop schema selection. Analyze errors on 2-hop questions and introduce schema-path actions instead of selecting individual relations independently. Add schema cue features and lexical action masking to reduce the action space to schema paths compatible with the question.

7. Evaluate performance. The final evaluation measures exact match and F1 on paraphrased biomedical questions, separately for 1-hop and 2-hop examples.

Challenging Variants

The main challenging variant is 2-hop schema selection.

In the 1-hop setting, the agent only needs to identify one Hetionet relation. This is relatively easier and reaches stronger performance. In the 2-hop setting, the agent must select a complete schema path composed of two relations. This is harder because the model must identify both relations and their correct combination.

The project therefore introduces schema-path actions, schema cue features, lexical action masking, and hierarchical hop-count selection. These mechanisms specifically target the 2-hop difficulty.

Further challenging extensions could include 3-hop schema selection, execution-based rewards using Neo4j, larger paraphrased datasets, and end-to-end Cypher generation.

Final Evaluation

Overall exact match: 0.45

Overall F1: 0.625

1-hop exact match: 0.70

1-hop F1: 0.767

2-hop exact match: 0.367

2-hop F1: 0.578